GTPA v2 — Block B: GTPA objects on the UFD base

Status:

```
document   = GTPA v2 / Block B answers / READ 2026-09-26 /
             one independent read found two real errors (fixed by the
             writer in 1e7081c, not re-read) and four minor points (applied)
folder     = gtpa-v2/ (text of the next GTPA edition + its crosswalk from v1.2)
base       = UFD nucleus v0.1, ufd/UFD-NUCLEO.md at research
             3c2052042aa16c7a07ea4be3a1d4bccc7428df45 (blob 4f52a62c):
             convention C1-C6, statements U-01..U-15, Block A answers
scope      = Block B of the approved architecture: EG-02.2, EG-02.4 (M_cog),
             T6, EG-02.5, EG-03.7, EG-04.1, EG-09.3, EG-10.b2
ceiling    = FORMAL_LOCAL_CONDITIONAL / NONPROBATIVE
claims     = NONE. Claim authority = gtpa-lab/CLAIM-LEDGER.md; nothing here
             is a GTPA claim until it passes the ledger gate (next free ID
             C0138; C0137 is held for EG-05c)
physics    = CLOSED     MATH<->GTPA = UNBOUND     MATH<->UFD = one object only
             (formal-framework/governance/math-owner-binding-record-n0-v1.md),
             not used here
producer   = Claude (Anthropic), workflow writing agent
evaluator  = one independent read, same provider (L-A1), 2026-09-26: FIX_NEEDED
             (Stratonovich bias sentence; compact-tube overgeneralization);
             both fixed, minor points applied
checks     = no human check; no cross-provider check; no formal (Lean) check
recompute  = decisive calculations recomputed by machine on 2026-09-26: 25 agree, 4 not mechanizable
```

Opening record. The 29 (+2) questions on GTPA v1.2 were opened by the operator on 2026-09-25 ("3, sí por supuesto"; architecture "Go"), in three blocks. Block A is answered in the UFD nucleus (UFD §3). This file answers Block B. Working style (operator, 2026-09-26): coherent text, one status per statement, one independent read; no tests, check modules or hash pins.

0. How to read this file

- Same format as the UFD nucleus (UFD §0). Every statement has an ID `B-nn`, its types and hypotheses, one status — `DECLARED`, `PROVED_HERE` (complete proof here, checkable by hand), `STANDARD_FACT` (exact locator) or `OPEN` — and one line in plain Spanish (`En llano`).
- Section 3 answers the eight questions. Each has the exact question text of the triage record (research `d0cf83e`), the GTPA v1.2 locators, exactly one verdict — `HOLDS_UNDER_STATED_HYPOTHESES` / `FAILS_WITH_COUNTEREXAMPLE` / `ILL_POSED_AS_WRITTEN` (+ the repair GTPA v2 adopts) / `OPEN` — the argument, the repair, and one sentence in plain Spanish. `CROSSWALK-v1.2.md` carries one row per answer.
- The base motion is UFD's: the first-order Itô SDE (M) on R^n and, under the pairing (P), its paired variant (M_P) (C1–C5; U-01, U-02, U-08). UFD's notation (UFD §0) is used unchanged; §1 adds what Block B needs.
- Words: UFD's rule applies. Words from physics appear only inside quotations or paraphrases of v1.2 and in explicit negations. v1.2's own names (M_{cog}, p_{est}, T_{info}, `Correction_Ito`, "impedance") are used as names of v1.2 objects and nothing more.
- Page numbers are PDF page indices of GTPA v1.2 (`quasar:modulos/formacion/fuentes/GTPA-v1.2.pdf`, SHA-256 $0c9434c1...865f$), the numbering of its text export (blob 32787369) and of the triage. Quotations were checked against the text export.
- No MATH object is used as proof anywhere in this file.

1. Notation added for Block B

- A perception map is a map $\Pi : R^n \to R^d$ of class C^2 or more, written in coordinates on both sides (v1.2: "Pi: M_base -> M_model", p.22). Π^a ($a = 1..d$) are its components, $D\Pi$ its $d \times n$ Jacobian, $D^2\Pi^a$ the symmetric $n \times n$ Hessian of Π^a.
- For a symmetric $n \times n$ matrix C, $\mathrm{Tr}_C\, D^2\Pi \in R^d$ has components $(\mathrm{Tr}_C\, D^2\Pi)^a := \sum_{i,j} C_{ij}\, \partial_i \partial_j\, \Pi^a = \mathrm{Tr}(C\, D^2\Pi^a)$. The two indices of C are paired with the two derivative (base) indices of Π^a; the model index a stays free. The other pairing gives the same number, since C and $D^2\Pi^a$ are symmetric; pairing a model index with a base index would need an identification of R^d with R^n that nothing supplies (for $d \neq n$ there is none).
- For an Itô SDE $dq = b\,dt + \Sigma\, dW$, the diffusion matrix is $\Sigma\Sigma^\mathsf{T}$, with $(\Sigma\Sigma^\mathsf{T})_{ij} = d\langle q^i, q^j\rangle_t/dt$ [Karatzas–Shreve 1991, §3.2]: the covariance of the noise increment per unit of time.
- A recoding of the model side is a C^2 diffeomorphism φ of R^d onto itself, acting by $\Pi \mapsto \varphi \circ \Pi$.
- $\|\cdot\|$ is the operator norm of a matrix, $\|\cdot\|_F$ the Frobenius norm.

2. Statements

B-01 — The Itô image of the base motion `[STANDARD_FACT + PROVED_HERE]` (EG-02.2, EG-03.7, EG-10.b2)

Let q solve $dq = b(q)\,dt + \Sigma(q)\,dW$ under (H) (U-01), and let $\Pi \in C^2(R^n; R^d)$.

(a) `STANDARD_FACT`. For every $t \geq 0$,

$$\Pi(q_t) = \Pi(q_0) + \int_0^t [D\Pi\, b + \tfrac{1}{2}\, \mathrm{Tr}_{\Sigma\Sigma^\mathsf{T}}\, D^2\Pi](q_s)\, ds + \int_0^t D\Pi(q_s)\, \Sigma(q_s)\, dW_s.$$

[Itô's formula, vector form: Øksendal 2003, Thm 4.2.1; U-05 (a) states it for J.] So the Itô drift of the image exceeds the push-forward $D\Pi\, b$ of the base drift by exactly $\tfrac{1}{2}\, \mathrm{Tr}_{\Sigma\Sigma^\mathsf{T}}\, D^2\Pi$, and the matrix that enters is $\Sigma\Sigma^\mathsf{T}$, the covariance of the noise increment per unit of time. Under (P), $\Sigma\Sigma^\mathsf{T} = 2\varepsilon G$ and the term is $\varepsilon\, \mathrm{Tr}_G\, D^2\Pi$.

(b) `PROVED_HERE` (Stratonovich reading). If the noise is read as $\Sigma \circ dW$, with $\Sigma \in C^2$, $\Pi \in C^3$ and Stratonovich drift $\tilde{b}$ (so $b = \tilde{b} + c_\Sigma$, U-02 (b)), then

$$\Pi(q_t) = \Pi(q_0) + \int_0^t D\Pi\, \tilde{b}\,(q_s)\, ds + \int_0^t (D\Pi\, \Sigma)(q_s) \circ dW_s :$$

the Stratonovich form of the image has no second-order term.

Proof. Put $Y^k := (D\Pi\, \Sigma_k)(q)$. Since $D\Pi\, \Sigma_k \in C^2$, Itô's formula makes Y^k a continuous semimartingale with martingale part $\sum_{l,k'} \int \partial_l (D\Pi\, \Sigma_k)(q)\, \Sigma_{lk'}(q)\, dW^{k'}$, so $d\langle Y^k, W^k\rangle_t = \sum_l \partial_l (D\Pi\, \Sigma_k)(q_t)\, \Sigma_{lk}(q_t)\, dt$, as in U-02 (b). Summing over k: $\tfrac{1}{2} \sum_k D(D\Pi\, \Sigma_k)\, \Sigma_k = \tfrac{1}{2} \sum_k [D^2\Pi(\Sigma_k, \Sigma_k) + D\Pi\, (D\Sigma_k)\, \Sigma_k] = \tfrac{1}{2}\, \mathrm{Tr}_{\Sigma\Sigma^\mathsf{T}}\, D^2\Pi + D\Pi\, c_\Sigma$. With $\int Y \circ dW^k = \int Y\, dW^k + \tfrac{1}{2}\langle Y, W^k\rangle$ this gives $\int D\Pi\Sigma \circ dW = \int D\Pi\Sigma\, dW + \int [\tfrac{1}{2}\, \mathrm{Tr}_{\Sigma\Sigma^\mathsf{T}}\, D^2\Pi + D\Pi\, c_\Sigma]\, ds$; substitute in (a) with $b = \tilde{b} + c_\Sigma$. ∎

So $\tfrac{1}{2}\, \mathrm{Tr}_{\Sigma\Sigma^\mathsf{T}}\, D^2\Pi$ is the Itô second-order term of Π. It is not the Itô–Stratonovich conversion term c_Σ of the base (UFD §3.2, item 3), and it does not appear in the Stratonovich description of the image.

En llano: vista a través de una lente curva Π, la deriva de Itô de lo percibido lleva, además de la deriva de la base empujada por la lente, un término $\tfrac{1}{2}\cdot\mathrm{Tr}(\Sigma\Sigma^\mathsf{T}\cdot \mathrm{D}^2\Pi)$ hecho de la curvatura de la lente y de la covarianza del ruido por unidad de tiempo; en la forma de Stratonovich ese término no aparece.

B-02 — Mean of the image versus image of the mean [PROVED_HERE] (EG-10.b2)

Let ξ be a random vector in R^n with $E|\xi|^3 < \infty$, mean m, covariance C, and $\delta := \xi - m$. Let $\Pi \in C^3(R^n; R^d)$ with $|D^3\Pi^a(x)[h,h,h]| \le K_3\,|h|^3$ for all x, h and every a. Then for every a

$$|\,E\,\Pi^a(\xi) - \Pi^a(m) - \tfrac{1}{2}\,(\mathrm{Tr}_C\, D^2\Pi)^a(m)\,| \le (K_3/6)\,E|\delta|^3.$$

Proof. Taylor's formula with Lagrange remainder for $t \mapsto \Pi^a(m + t\delta)$ on $[0,1]$: $\Pi^a(\xi) = \Pi^a(m) + D\Pi^a(m)\delta + \tfrac{1}{2}\,\delta^{\mathsf{T}} D^2\Pi^a(m)\delta + R$ with $|R| \le K_3|\delta|^3/6$. Take expectations: $E[D\Pi^a(m)\delta] = 0$ and $E[\delta^{\mathsf{T}} H \delta] = \mathrm{Tr}(H\,E[\delta\delta^{\mathsf{T}}]) = \mathrm{Tr}(HC)$. ∎

So, to second order in the noise, the gap named on p.38 ("Mean(Pi(Noise)) != Pi(Mean(Noise))") is $+\tfrac{1}{2}\,\mathrm{Tr}_C\, D^2\Pi(m)$: Thm 6.1's expression with the opposite sign. If $\delta = s \cdot \zeta$ for a fixed law of ζ, the gap is of order s^2 and the remainder of order s^3.

En llano: la media de lo que se ve a través de una lente curva no es la imagen de la media; la diferencia es ½·Tr(C·D²Π), salvo un error del tamaño del tercer momento del ruido.

B-03 — The curvature term under a recoding of the model side [PROVED_HERE] (EG-10.b2)

Let $\Pi \in C^2(R^n; R^d)$, φ a C^2 diffeomorphism of R^d, C symmetric. At every q, with $y = \Pi(q)$,

$$\mathrm{Tr}_C\, D^2(\varphi \circ \Pi)(q) = D\varphi(y) \cdot \mathrm{Tr}_C\, D^2\Pi(q) + \mathrm{Tr}_{D\Pi(q)\,C\,D\Pi(q)^{\mathsf{T}}}\, D^2\varphi(y).$$

Proof. Chain rule twice: $\partial_i\partial_j(\varphi^k \circ \Pi) = \sum_a \partial_a\varphi^k\,\partial_i\partial_j\Pi^a + \sum_{a,b} \partial_a\partial_b\varphi^k\,\partial_i\Pi^a\,\partial_j\Pi^b$; contracted with C_{ij}, the second sum is $\sum_{a,b} \partial_a\partial_b\varphi^k\,(D\Pi\,C\,D\Pi^{\mathsf{T}})_{ab}$. ∎

Consequences for the base motion of B-01 (a), with $C = \Sigma\Sigma^{\mathsf{T}}$: (a) the push-forward of the base drift transforms as a vector: $D(\varphi \circ \Pi)\,b = D\varphi\,(D\Pi\,b)$, exactly; (b) the Itô drift of the image transforms as $\mathrm{drift}(\varphi \circ \Pi) = D\varphi \cdot \mathrm{drift}(\Pi) + \tfrac{1}{2}\,\mathrm{Tr}_{D\Pi\,\Sigma\Sigma^{\mathsf{T}}\,D\Pi^{\mathsf{T}}}\, D^2\varphi$: a vector law plus a term made of the curvature of the recoding φ, not of Π; (c) Thm 6.1's expression for the recoded map, $-\tfrac{1}{2}\,\mathrm{Tr}_{\Sigma\Sigma^{\mathsf{T}}}\, D^2(\varphi \circ \Pi)$, equals $D\varphi \cdot (-\tfrac{1}{2}\,\mathrm{Tr}_{\Sigma\Sigma^{\mathsf{T}}}\, D^2\Pi) - \tfrac{1}{2}\,\mathrm{Tr}_{D\Pi\,\Sigma\Sigma^{\mathsf{T}}\,D\Pi^{\mathsf{T}}}\, D^2\varphi$.

En llano: si se cambian las coordenadas del lado del modelo, el término de la lente no se transforma como un vector: aparece un término nuevo que sale de la curvatura del propio cambio de coordenadas; la deriva empujada sí se transforma como un vector.

B-04 — What does not depend on the model-side coordinates [PROVED_HERE] (EG-10.b2)

Two equivalences on presentations: $\sim_{\mathrm{diff}}$, $\Pi \sim \varphi \circ \Pi$ for C^2 diffeomorphisms φ of R^d; $\sim_{\mathrm{aff}}$, the same with φ affine ($\varphi(y) = Ly + c$, L invertible). GTPA-C0121 calls a label representation invariant when it is constant on classes, i.e. when it descends to the quotient. For a point- or vector-valued quantity attached to Π this means that it names one point (one tangent vector) of the model side whatever its coordinates, i.e. that it is equivariant: its value for $\varphi \circ \Pi$ is φ (resp. $D\varphi$) applied to its value for Π.

(a) Drift level (the base motion of B-01 (a)). The corrected drift $D\Pi\,b$ — the Itô drift of the image minus $\tfrac{1}{2}\,\mathrm{Tr}_{\Sigma\Sigma^{\mathsf{T}}}\, D^2\Pi$ — is equivariant under $\sim_{\mathrm{diff}}$ (B-03 (a)): it descends exactly. The uncorrected Itô drift of the image is equivariant under $\sim_{\mathrm{aff}}$ (there $D^2\varphi = 0$) but not under $\sim_{\mathrm{diff}}$: for $n = d = 1$, $\Pi = \mathrm{id}$, $b = 0$, $\Sigma = \sigma > 0$ and $\varphi(y) = y + y^3/3$, the image drift is 0 for Π and $\tfrac{1}{2}\sigma^2\varphi''(q) = \sigma^2 q$ for $\varphi \circ \Pi$, while $D\varphi \cdot 0 = 0$.

(b) Static level (the setting of B-02). The corrected centroid $c(\Pi) := E\,\Pi(\xi) - \tfrac{1}{2}\,\mathrm{Tr}_C\, D^2\Pi(m)$ satisfies $c(\varphi \circ \Pi) = \varphi(c(\Pi))$ exactly for affine φ. For a C^3 diffeomorphism φ with $\sup \|D\varphi\| < \infty$, and Π, $\varphi \circ \Pi$ both satisfying the third-derivative bound of B-02, $|c(\varphi \circ \Pi) - \varphi(c(\Pi))| \le K\,E|\delta|^3$, with K depending

only on those bounds, on d and on $\sup\|D\varphi\|$. The defect can be nonzero: for $n = d = 1$, $\Pi = \mathrm{id}$, $\varphi(y) = y + y^3/3$ and $E\,\delta^3 = \kappa \neq 0$, $c(\varphi) - \varphi(c(\mathrm{id})) = \kappa/3$. The point $\Pi(m)$ is equivariant under $\sim_{\mathrm{diff}}$ exactly: $\varphi(\Pi(m)) = (\varphi \circ \Pi)(m)$.

Proof of (b). Affine φ: $D^2(\varphi \circ \Pi) = L\,D^2\Pi$, so $c(\varphi \circ \Pi) = L\,E\,\Pi(\xi) + c - \tfrac{1}{2} L\ \mathrm{Tr}_C\, D^2\Pi(m) = \varphi(c(\Pi))$. General φ: B-02 bounds $|c(\Pi) - \Pi(m)|$ and $|c(\varphi \circ \Pi) - \varphi(\Pi(m))|$ by multiples of $E|\delta|^3$; $|\varphi(c(\Pi)) - \varphi(\Pi(m))| \le \sup\|D\varphi\| \cdot |c(\Pi) - \Pi(m)|$; the triangle inequality concludes. Example: $E\,\varphi(\xi) = m + E(m+\delta)^3/3 = \varphi(m) + mC + \kappa/3$ and $\tfrac{1}{2}C\varphi''(m) = mC$, so $c(\varphi) = \varphi(m) + \kappa/3$, while $c(\mathrm{id}) = m$. ∎

En llano: lo que no depende de las coordenadas del modelo es la deriva corregida (exactamente) y la imagen de la media; la media corregida lo es exactamente para cambios afines y, para cambios curvos, salvo un error de tercer orden en el ruido.

B-05 — A term in the p_est equation acts one order later in time [PROVED_HERE] (EG-03.7)

Consider the linear system with the shape of Ax 6.1 and constant matrices: $\dot q_{\mathrm{est}} = M^{-1}\,p_{\mathrm{est}}$, $\dot p_{\mathrm{est}} = -\Gamma_{\mathrm{cog}}\,p_{\mathrm{est}} + F(t) + \kappa(t)$, where M is symmetric positive definite, Γ_{cog} is a constant $d \times d$ matrix and F, κ are continuous. By linearity, the contribution of κ to q_{est} is

$$\delta q(t) = M^{-1} \int_0^t \int_0^s e^{-\Gamma_{\mathrm{cog}}\,(s-r)}\,\kappa(r)\,dr\,ds.$$

For constant κ: $\delta q(t) = \tfrac{1}{2} M^{-1}\,\kappa\,t^2 + O(t^3)$ as $t \to 0$; and if every eigenvalue of Γ_{cog} has positive real part, $\delta q(t)/t \to M^{-1}\,\Gamma_{\mathrm{cog}}^{-1}\,\kappa$ as $t \to \infty$. The same κ placed in the q_{est} equation ($\dot q_{\mathrm{est}} = M^{-1}\,p_{\mathrm{est}} + \kappa$) contributes exactly $\kappa\,t$.

Proof. Variation of constants gives the display. For constant κ, $\int_0^s e^{-\Gamma_{\mathrm{cog}}\,u}\,du\,\kappa = s\kappa + O(s^2)$; integrate once more. If the spectrum of Γ_{cog} lies in $\{\mathrm{Re} > 0\}$, Γ_{cog} is invertible and $\int_0^s e^{-\Gamma_{\mathrm{cog}}\,u}\,du\,\kappa = \Gamma_{\mathrm{cog}}^{-1}(I - e^{-\Gamma_{\mathrm{cog}}\,s})\,\kappa \to \Gamma_{\mathrm{cog}}^{-1}\,\kappa$; the time average of a convergent function converges to its limit. ∎

So a constant term κ in the p_{est} equation offsets a constant rate β of the q_{est} coordinate only in the long run, and only if $M^{-1}\,\Gamma_{\mathrm{cog}}^{-1}\,\kappa = -\beta$; the choice $\kappa = -\beta$ does so for every β only when $\Gamma_{\mathrm{cog}}\,M = I$. At short times it acts at order t^2 against a bias of order t.

En llano: un término sumado a la ecuación de p_est solo mueve la estimación a orden t² (no t), y a la larga el término −β solo compensa un desvío constante β en el ritmo de q_est si Γ_cog·M_cog es la identidad; puesto en la ecuación de q_est lo compensa exactamente.

B-06 — A witness for Thm 6.1 [PROVED_HERE] (EG-03.7)

$n = d = m = 2$. Base: the paired motion (M_P) with $J \equiv 0$, $A = 0$, $G = I$, $\Sigma = \sqrt{(2\varepsilon)}\,I$, so $q_t = q_0 + \sqrt{(2\varepsilon)}\,W_t$ (deterministic q_0). Let $\Pi(q) := (q_1, q_2 + q_1^2/2)$: a C^∞ diffeomorphism of R^2 onto R^2 ($\det D\Pi = 1$, inverse $(y_1, y_2 - y_1^2/2)$), so it meets Ax 2.2 ("det(Jacobian(Pi)) != 0 everywhere", p.22). Then $\tfrac{1}{2}\,\mathrm{Tr}_{\Sigma\Sigma^{\mathsf T}}\,D^2\Pi = (0, \varepsilon) =: \bar\kappa$ is constant, and for all $t \ge 0$

$$E[\Pi(q_t)] = \Pi(q_0) + \bar\kappa\,t, \qquad \Pi(E[q_t]) = \Pi(q_0).$$

So the perceived mean leaves the image of the mean at the constant rate $\bar\kappa \neq 0$. With constant M_{cog} and Γ_{cog}, Thm 6.1's term $-\bar\kappa$ placed in the p_{est} equation moves q_{est} by $-\tfrac{1}{2} M_{\mathrm{cog}}^{-1}\,\bar\kappa\,t^2 + O(t^3)$ (B-05), which does not offset $\bar\kappa\,t$ at small t; placed in the q_{est} equation it moves it by exactly $-\bar\kappa\,t$, and $E[\Pi(q_t)] - \bar\kappa\,t = \Pi(E[q_t])$ for all t.

Proof. $D^2\Pi^1 = 0$ and $D^2\Pi^2 = \mathrm{diag}(1, 0)$, so $\tfrac{1}{2}\,\mathrm{Tr}(2\varepsilon I \cdot D^2\Pi^2) = \varepsilon$. $E\,q_t = q_0$ and $E[q_{1,t}^2] = q_{0,1}^2 + 2\varepsilon t$, hence $E\,\Pi^2(q_t) = q_{0,2} + \tfrac{1}{2}\,q_{0,1}^2 + \varepsilon t$. ∎

En llano: ejemplo comprobable a mano: con una lente que se curva en una sola dirección y ruido constante, la media de lo percibido se separa de la imagen de la media a ritmo constante ε; la corrección de Thm 6.1 puesta en la ecuación de p_est no lo compensa a tiempos cortos, y puesta en la de q_est sí, exactamente.

B-07 — The matrix (H² + δI)^{1/2} [STANDARD_FACT + PROVED_HERE] (EG-02.4, T6)

Let H be a real symmetric $n \times n$ matrix, $H = U\ \mathrm{diag}(\lambda_i)\, U^{\mathsf{T}}$ with U orthogonal, and $\delta > 0$. (δ is v1.2's "epsilon" / "epsilon_inertial", renamed because ε is UFD's noise level, C4.)

(a) STANDARD_FACT. $H^2 + \delta I$ is positive definite and has a unique symmetric positive definite square root $M(H) := (H^2 + \delta I)^{1/2}$ [Horn & Johnson 2013, Thm 7.2.6]; by uniqueness, $M(H) = U\ \mathrm{diag}((\lambda_i^2 + \delta)^{1/2})\, U^{\mathsf{T}}$.

(b) PROVED_HERE. $M(H) \succeq \sqrt{\delta}\, I$, so $M(H)$ is invertible and $\|M(H)^{-1}\| \leq \delta^{-1/2}$. Hence v1.2's "Kinetic Stability: The Cognitive Mass M_cog is positive definite" (Thm 5.1, condition 2, p.36) holds automatically for the object of Def 1.1; it is not a condition.

(c) PROVED_HERE. $M(-H) = M(H)$. So $M(H)$ does not distinguish a non-degenerate minimum of J from a maximum, nor from a saddle with the same absolute eigenvalues; the signed Hessian does (U-12 (b), (c)). This matches Def 1.1's own wording: its "Valley Regime" and its "Saddle Regime" are both "stiff" (p.18).

(d) PROVED_HERE. The spectrum of H is not a function of $M(H)$: for $n = 1$, $H = 1$ and $H = -1$ give the same M. Nor is $M(H)$ a function of the spectrum of H: for $n = 2$, $H_1 = \mathrm{diag}(1, 2)$ and $H_2 = R\, H_1\, R^{\mathsf{T}}$, with R the rotation by $\pi/4$, have the same spectrum, but $M(H_2) = R\, M(H_1)\, R^{\mathsf{T}} \neq M(H_1)$, because $M(H_1) = \mathrm{diag}((1+\delta)^{1/2}, (4+\delta)^{1/2})$ has distinct diagonal entries and so commutes only with diagonal matrices. Both are functions of H.

Proofs. (b) The eigenvalues $(\lambda_i^2+\delta)^{1/2}$ are $\geq \sqrt{\delta}$. (c) $(-H)^2 = H^2$. (d) $M(R\, H_1\, R^{\mathsf{T}}) = (R(H_1^2+\delta I)R^{\mathsf{T}})^{1/2} = R\, M(H_1)\, R^{\mathsf{T}}$ by the uniqueness in (a). ∎

En llano: la "masa" de la Def 1.1 es una matriz bien definida y siempre invertible, pero borra el signo de la curvatura (no distingue un valle de un punto de silla), y ni su espectro es el de la Hessiana ni al revés.

B-08 — Frequency response of linear motions [PROVED_HERE] (EG-04.1)

(a) Second order, one mode. Let $m > 0$, $c \geq 0$, $k > 0$, $\omega > 0$, $F \in C \setminus \{0\}$ and $\omega_0 := (k/m)^{1/2}$. If $c > 0$ or $\omega \neq \omega_0$, the equation $m\,\ddot{x} + c\,\dot{x} + k\,x = \mathrm{Re}(F\, e^{i\omega t})$ has exactly one solution of the form $\mathrm{Re}(X\, e^{i\omega t})$, namely $X = F/(k - m\omega^2 + ic\omega)$. Define the ratios $R(\omega) := F/(i\omega X) = c + i(m\omega - k/\omega)$ (forcing per unit of time derivative) and $K(\omega) := F/X = k - m\omega^2 + ic\omega$ (forcing per unit of displacement). Then:

- $|R(\omega)| \geq c$, with equality iff $\omega = \omega_0$: the minimum of $|R|$ is c, which is zero iff $c = 0$;
- if $c > 0$ and $c^2 < 2mk$, $|K|$ on $\omega > 0$ is minimized exactly at $\omega_* = (\omega_0^2 - c^2/(2m^2))^{1/2} < \omega_0$, with $|K(\omega_*)| = c\,(\omega_0^2 - c^2/(4m^2))^{1/2} > 0$; if $c > 0$ and $c^2 \geq 2mk$, $|K|$ is strictly increasing on $\omega > 0$.

Proof. Substituting $\mathrm{Re}(X\, e^{i\omega t})$ gives $\mathrm{Re}((k - m\omega^2 + ic\omega)\, X\, e^{i\omega t}) = \mathrm{Re}(F\, e^{i\omega t})$ for all t, i.e. $(k - m\omega^2 + ic\omega)\, X = F$; the coefficient vanishes iff $c = 0$ and $\omega = \omega_0$. $|R|^2 = c^2 + (m\omega - k/\omega)^2$, and $m\omega = k/\omega$ iff $\omega = \omega_0$. $|K|^2 = f(\omega^2)$ with $f(s) = (k - ms)^2 + c^2 s$, a convex quadratic with $f'(s) = 2m^2 s - 2mk + c^2$; its minimizer $s_* = \omega_0^2 - c^2/(2m^2)$ is positive iff $c^2 < 2mk$; then $k - m\,s_* = c^2/(2m)$ and $f(s_*) = c^2(\omega_0^2 - c^2/(4m^2)) > 0$. If $c^2 \geq 2mk$, $f' > 0$ on $s > 0$. ∎

(b) Second order, several modes. Let $M \succ 0$ and $H \succ 0$ be symmetric $n \times n$ and $C = \gamma M$ with $\gamma \geq 0$ — the shape of v1.2's (q,p) forms, where the term "- Gamma * p" with $p = \dot{M}x$ gives $C = \Gamma M$ for a scalar

Γ. Let U be orthogonal with $M^{-1/2} H M^{-1/2} = U \operatorname{diag}(\omega_j^2) U^\top$ and $\Phi := M^{-1/2} U$. Then $\Phi^\top M \Phi = I$, $\Phi^\top H \Phi = \operatorname{diag}(\omega_j^2)$, $\Phi^\top C \Phi = \gamma I$, and $x = \Phi y$ turns $M\ddot{x} + C\dot{x} + Hx = f$ into $\ddot{y}_j + \gamma\, \dot{y}_j + \omega_j^2\, y_j = (\Phi^\top f)_j$: n copies of (a) with $m = 1$, $c = \gamma$, $k = \omega_j^2$. The ω_j^2 are the eigenvalues of $M^{-1}H$: they depend on M, not on H alone. (Direct computation.)

(c) First order: UFD's base, noiseless, linearized at a non-degenerate minimum q^* of J, with $G^* := G(q^*) \succ 0$, $H^* := D^2 J(q^*) \succ 0$, $A^* := A(q^*)$. Since $\nabla J(q^*) = 0$, the linearization of $\dot{q} = (A - G)\nabla J$ is $\dot{x} = (A^* - G^*) H^* x$; add a forcing f. If $A^* = 0$, $G^* H^*$ is similar to ${G^*}^{1/2} H^* {G^*}^{1/2} \succ 0$, whose eigenvalues μ_j are positive; in the corresponding coordinates each mode is $\dot{x}_j = -\mu_j x_j + f_j$, with ratios $F/X = \mu_j + i\omega$ (modulus increasing on $\omega \geq 0$) and $F/(i\omega X) = 1 - i\mu_j/\omega$ (modulus decreasing on $\omega > 0$): neither has a minimum at a positive frequency. A circulating part can create one, and there the minimum is again the sliding coefficient: for $n = 2$, $G^* = gI$, $A^* = aS$ ($S = [[0, 1], [-1, 0]]$), $H^* = kI$, $g, k > 0$, $a \neq 0$, the complex coordinate $z = x_1 + i\, x_2$ turns the linearization into $\dot{z} = -k(g + ia)\, z + f_z$; for $f_z = F\, e^{i\omega t}$ the ratio is $kg + i(\omega + ka)$, whose modulus is minimized at $\omega = -ka$ (a positive frequency when the forcing turns like the free motion e^{-ikat}), with minimum $kg > 0$.

(d) Def 7.2's $Z = B^\top G_{\text{reg}}\, B + \lambda_{\text{effort}}\, I$ is the Hessian in u of the quadratic `Action_Local` (p.44): a constant matrix with no frequency argument. If $G_{\text{reg}} \succeq 0$ (as its use as a norm requires) and $\lambda_{\text{effort}} > 0$, then $Z \succeq \lambda_{\text{effort}}\, I \succ 0$: it never vanishes, and it is neither $R(\omega)$ nor $K(\omega)$ of any motion. (Direct.)

En llano: en un sistema lineal de segundo orden, a la frecuencia propia de cada modo desaparece la parte "reactiva" y queda la amortiguación, que solo es cero si no hay amortiguación; el movimiento de primer orden de UFD no tiene frecuencia preferida salvo que haya una parte que gire; y la Z de la Def 7.2 es otra cosa: una matriz fija que nunca es cero.

B-09 — A noise level that changes in time `[PROVED_HERE]` (EG-02.5)

$n = m = 1$, $J(x) = x^2/2$, $A = 0$, $G = 1$, and the pairing imposed at every time: $\sigma(t)^2 = 2\varepsilon(t)$ with $\varepsilon \in C^1([0, T])$, $\varepsilon > 0$ (the paired drift is $-x$; $\operatorname{div} G = 0$). Let x_0 be F_0-measurable with mean 0 and variance $\varepsilon(0)$, and let x solve the linear equation $dx = -x\, dt + \sigma(t)\, dW$. Then $E\, x_t = 0$ and

$$\operatorname{Var} x_t = \varepsilon(t) - \int_0^t e^{-2(t-s)}\, \varepsilon'(s)\, ds, \qquad 0 \leq t \leq T.$$

Hence $\operatorname{Var} x_t = \varepsilon(t)$ — the variance of the instantaneous law $\propto e^{-J/\varepsilon(t)}$, i.e. $N(0, \varepsilon(t))$ — for every $t \in [0, T]$ iff ε is constant on $[0, T]$; and if $\varepsilon'(0) \neq 0$ the law leaves that family at once: $\operatorname{Var} x_t - \varepsilon(t) = -\varepsilon'(0)\, t + o(t)$.

Proof. By Itô's product rule (as in U-06), $d(e^t\, x_t) = e^t\, \sigma(t)\, dW_t$, so $x_t = e^{-t}\, x_0 + e^{-t} \int_0^t e^s\, \sigma(s)\, dW_s$; this is the only solution, since the difference of two solutions solves $\dot{d} = -d$, $d(0) = 0$. The integral $\int_0^t e^s\, \sigma(s)\, dW_s$ is a square-integrable martingale started at 0, so it has conditional mean 0 given F_0 and is uncorrelated with x_0; by the Itô isometry [Øksendal 2003, Cor. 3.1.7], $\operatorname{Var} x_t = e^{-2t}\, \varepsilon(0) + \int_0^t 2e^{-2(t-s)}\, \varepsilon(s)\, ds$. Integration by parts: $\int_0^t 2e^{-2(t-s)}\, \varepsilon(s)\, ds = \varepsilon(t) - e^{-2t}\, \varepsilon(0) - \int_0^t e^{-2(t-s)}\, \varepsilon'(s)\, ds$. If $\operatorname{Var} x_t = \varepsilon(t)$ for all t, then $\int_0^t e^{2s}\, \varepsilon'(s)\, ds = 0$ for all t, so $\varepsilon' \equiv 0$; the converse is read off the formula. ∎

En llano: si el nivel de ruido cambia con el tiempo, se puede imponer el emparejamiento en cada instante, pero el reparto no sigue al "equilibrio del instante": va con retraso, y solo coincide con él en todo momento si el nivel es constante.

B-10 — Signed efficacy of a drift lift on the base motion [PROVED_HERE; (b) conditional on (K)] (EG-09.3)

Setting (L). Integers $n, m \geq 1$, $\bar{\eta} > 0$; $f0, g : R^n \to R^n$ and $\Sigma : R^n \to R^{n\times m}$ locally Lipschitz; for every $\eta \in [0, \bar{\eta}]$ the Itô SDE

$$(L_\eta) \quad dx = (f0 + \eta\, g)(x)\, dt + \Sigma(x)\, dW, \qquad x(0) = x_0 \in R^n \text{ (deterministic)},$$

satisfies (H) (U-01); x^η denotes its solution. The base drift $f0$ is the drift of (M) or of (M_P) (for instance $f0 = b_P$); ηg is the lift. For $f \in C^2$, $L_\eta f := \langle f0 + \eta g, \nabla f\rangle + \tfrac{1}{2}\operatorname{Tr}(\Sigma\Sigma^\top D^2 f) = L_0 f + \eta\langle g, \nabla f\rangle$. The observable $V \in C^2(R^n)$ is lower-is-better, and the signed efficacy, with the sign convention of GTPA-C0115, is $\Gamma_T := E\,V(x_T^0) - E\,V(x_T^\eta)$ (assumed finite).

(a) Exact ablation: at $\eta = 0$, (L_η) is the base equation and its solution is x^0.

(b) Representation. Hypothesis (K): the function $u(t,x) := E\,V(x_t^{0,x})$, where $x^{0,x}$ solves (L_0) from x, belongs to $C^{1,2}([0,T]\times R^n)$ and satisfies $\partial_t u = L_0 u$ there with $u(0,\cdot) = V$; and $E\int_0^T |\Sigma^\top \nabla u(T-s, x_s^\eta)|^2\, ds < \infty$, $E\int_0^T |\langle g, \nabla u(T-s,\cdot)\rangle(x_s^\eta)|\, ds < \infty$. Under (K),

$$(\bigstar) \quad \Gamma_T = -\eta\, E\int_0^T \langle g, \nabla u(T-s,\cdot)\rangle\,(x_s^\eta)\, ds.$$

Proof. Apply Itô's rule for $C^{1,2}$ functions [Karatzas–Shreve 1991, Thm 3.3.6] to $s \mapsto u(T-s, x_s^\eta)$ on $[0, T]$: its drift is $(-\partial_t u + L_\eta u)(T-s, x_s^\eta) = \eta\,\langle g, \nabla u(T-s,\cdot)\rangle(x_s^\eta)$, and its martingale part $\int \nabla u(T-s, x_s^\eta)^\top \Sigma(x_s^\eta)\, dW_s$ has mean 0 by (K) [Øksendal 2003, Thm 3.2.1]. Taking expectations between $s = 0$ and $s = T$: $E\,V(x_T^\eta) - u(T, x_0) = \eta\, E\int_0^T \langle g, \nabla u(T-s,\cdot)\rangle(x_s^\eta)\, ds$, and $u(T, x_0) = E\,V(x_T^0)$. ∎

(K) is a named hypothesis, like (R) in U-07 (b); checking it for given coefficients is a separate task, not done here (it is not vacuous: its analogue holds in the explicit instances of B-11 (f)). Without noise, $u(t,x) = V(\varphi_t(x))$ for the base flow φ_t, and (★) is the exact identity obtained by differentiating $s \mapsto V(\varphi_{T-s}(x_s^\eta))$, the deterministic form behind GTPA-C0115's adjoint representation. In general (★) is Duhamel's formula for L_0 and L_η with the unbounded perturbation $\langle g, \nabla\cdot\rangle$, a case GTPA-C0112 (bounded perturbations) does not cover.

(c) Bound and first variation. Assume moreover that $f0$ and Σ are globally Lipschitz with constant L (Frobenius norm for Σ), that g is globally Lipschitz with $|g| \leq K_g$, and drive both arms by the same W (Γ_T depends only on the two laws). Then for $0 \leq t \leq T$

$$E\,|\,x_t^\eta - x_t^0\,|^2 \leq 3\,\eta^2\,K_g^2\,t^2\,\exp(3\,L^2\,(t+1)\,t),$$

so $|\Gamma_T| \leq \sqrt{3}\,L_V\,K_g\,\eta\,T\,\exp(3L^2(T+1)T/2)$ when V is Lipschitz with constant L_V. If also (K) holds for every $\eta \in (0, \bar{\eta}]$ and $h(s,x) := \langle g(x), \nabla u(T-s, x)\rangle$ is bounded and continuous on $[0,T]\times R^n$, then, with $P_s^0 f(x) := E\,f(x_s^{0,x})$,

$$\lim_{\eta\to 0} \Gamma_T/\eta = -E\int_0^T h(s, x_s^0)\, ds = -\int_0^T P_s^0(\langle g, \nabla P_{T-s}^0\, V\rangle)(x_0)\, ds.$$

Proof. $e_t := x_t^\eta - x_t^0 = \int_0^t [f0(x^\eta) - f0(x^0)]\, ds + \eta\int_0^t g(x^\eta)\, ds + \int_0^t [\Sigma(x^\eta) - \Sigma(x^0)]\, dW$. By $|a+b+c|^2 \leq 3(|a|^2 + |b|^2 + |c|^2)$, Cauchy–Schwarz in ds and the Itô isometry, $\psi(t) := E|e_t|^2$ satisfies $\psi(t) \leq 3\eta^2K_g^2t^2 + 3L^2(t+1)\int_0^t \psi(s)\, ds$; ψ is finite and bounded on $[0,T]$ [Karatzas–Shreve 1991, Thm 5.2.9], and Gronwall's lemma on $[0,t]$ gives the bound. Then $|\Gamma_T| \leq L_V\,E|e_T| \leq L_V\,\psi(T)^{1/2}$. For the limit, divide (★) by η: for each s, $x_s^\eta \to x_s^0$ in L^2, hence in probability, so $E\,h(s, x_s^\eta) \to E\,h(s, x_s^0)$ (bounded continuous h; pass to almost surely convergent subsequences); dominated convergence in s concludes. ∎

(d) Short-time limit. If $L_0 V$ and $\langle g, \nabla V\rangle$ are bounded and continuous and $E\int_0^{T_0} |\Sigma^\top \nabla V(x_s^\eta)|^2\, ds < \infty$ for some $T_0 > 0$ and each $\eta \in [0, \bar{\eta}]$, then

$$\lim_{T\to 0} \Gamma_T/T = -\eta\,\langle g(x_0), \nabla V(x_0)\rangle,$$

the same limit as without noise: the noise term ½ $\mathrm{Tr}(\Sigma\Sigma^{\mathsf{T}} D^2 V)$ is common to both arms and cancels.

Proof. For $T \le T_0$, Itô's formula and the zero mean of the stochastic integral give $E\,V(x_T^\eta) = V(x_0) + E\int_0^T L_\eta\,V(x_s^\eta)\,ds$, and the same at $\eta = 0$. For a bounded continuous F and a continuous path started at x_0, $T^{-1}\int_0^T F(x_s)\,ds \to F(x_0)$; dominated convergence passes this to the expectations. With $F = L_0\,V$ and $F = L_\eta\,V$: $\Gamma_T/T \to (L_0\,V - L_\eta\,V)(x_0) = -\eta\,\langle g, \nabla V\rangle(x_0)$. ∎

(e) Change of coordinates. Let φ be a C^3 diffeomorphism of R^n onto R^n. Then $y^\eta := \varphi(x^\eta)$ solves

$$dy = [\tilde b_0 + \eta\,\varphi_* g](y)\,dt + \tilde\Sigma(y)\,dW,$$
$$\tilde b_0 := [D\varphi\, f0 + \tfrac{1}{2}\,\mathrm{Tr}_{\Sigma\Sigma^{\mathsf{T}}} D^2\varphi] \circ \varphi^{-1}, \quad \varphi_* g := (D\varphi\, g)\circ\varphi^{-1}, \quad \tilde\Sigma := (D\varphi\,\Sigma)\circ\varphi^{-1},$$

whose coefficients are locally Lipschitz, so y^η is its unique solution. The transported family again has the form (L_η), with exact ablation and lift direction $\varphi_* g$; the Itô term of φ enters the base drift only. With $\tilde V := V\circ\varphi^{-1}$, $\tilde V(y_T^\eta) = V(x_T^\eta)$ on every path, so Γ_T is the same number in both presentations. Proof: B-01 (a) with $\Pi = \varphi$. ∎

(f) Stratonovich input. If the SDE is given as $dx = (f0 + \eta g)\,dt + \Sigma \circ dW$ with $\Sigma \in C^2$ not depending on η, its Itô form (U-02 (b)) is (L_η) with $f0$ replaced by $f0 + c_\Sigma$ and the same g: the conversion term is common to both arms and the ablation stays exact.

(g) No compact tube. For $n = m = 1$, $f0 = g = 0$, $\Sigma = \sigma > 0$, $x_T = x_0 + \sigma W_T$ has law $N(x_0, \sigma^2 T)$, which gives positive mass to the complement of every compact set, for every $T > 0$. So for this example no compact tube contains the paths almost surely, and a compact tube cannot be assumed in general; (a)–(e) use global hypotheses on R^n instead. This is not a claim about every diffusion with nonzero noise: noise that vanishes at the boundary can keep the paths in a compact set (for instance $n = m = 1$, zero Itô drift, $\Sigma(x) = \sigma(1 - x^2)$ near $[-1, 1]$, $x_0 \in (-1, 1)$; ± 1 is then never reached, by Feller's test, a `STANDARD_FACT`: Karatzas–Shreve, *Brownian Motion and Stochastic Calculus*, 2nd ed., §5.5.C, Thm 5.29). (A compact state space without boundary is another option, not worked out here.)

En llano: con ruido, la eficacia media de una acción que se suma a la deriva se escribe exactamente con la ecuación de Kolmogorov hacia atrás, y siguen valiendo su primera variación, su límite a tiempos cortos (el mismo que sin ruido, porque el ruido común se cancela) y su independencia de las coordenadas; lo que deja de valer es el "tubo compacto" del caso sin ruido, porque el ruido puede sacar las trayectorias de cualquier compacto (lo hace, por ejemplo, con ruido que no se anula en ningún punto).

B-11 — A lift that acts through the paired mobility `[PROVED_HERE; parts conditional as stated]` (EG-09.3)

Setting. $\varepsilon > 0$; $J \in C^2$; A as in (P) (C^1, antisymmetric, $\mathrm{div}_c\,A = 0$); G, G_1 symmetric matrix fields of class C^2 with $G_\eta := G + \eta G_1 \succeq 0$ for $\eta \in [0, \bar\eta]$; for each η a noise amplitude Σ_η with $\Sigma_\eta\,\Sigma_\eta^{\mathsf{T}} = 2\varepsilon G_\eta$ is fixed so that the paired motion (M_P) with data (A, G_η, ε),

$$dx = [(A - G_\eta)\nabla J + \varepsilon\,\mathrm{div}\,G_\eta](x)\,dt + \Sigma_\eta(x)\,dW, \qquad x(0) = x_0,$$

satisfies (H); x^η is its solution and Γ_T is defined as in B-10. The pairing is kept in both arms, so the lift changes the noise as well.

(a) $L_\eta = L_0 + \eta A_1$, where for $f \in C^2$

$$A_1 f = \langle -G_1\nabla J + \varepsilon\,\mathrm{div}\,G_1, \nabla f\rangle + \varepsilon\,\mathrm{Tr}(G_1 D^2 f) = \varepsilon\, e^{J/\varepsilon}\,\mathrm{div}(e^{-J/\varepsilon}\,G_1\,\nabla f).$$

The lift operator is of second order. Proof. The generator of (M_P) is $Lf = \langle b_P, \nabla f\rangle + \varepsilon\,\mathrm{Tr}(G\,D^2 f)$ (U-08); subtract the two arms. For the second form, $\mathrm{div}(\rho\,G_1\nabla f) = \langle\nabla\rho, G_1\nabla f\rangle + \rho\langle\mathrm{div}\,G_1, \nabla f\rangle +$

$\rho\ \mathrm{Tr}(G_1 D^2 f)$ (for symmetric G_1 the divergence of its columns is $\mathrm{div}\, G_1$), and $\nabla \rho_J = -\rho_J \nabla J/\varepsilon$ for $\rho_J = e^{-J/\varepsilon}$. ∎

(b) With $\langle g, \nabla \cdot \rangle$ replaced by A_1, the proofs of B-10 (b) and (d) apply unchanged: under the analogue of (K) (with Σ_η and $A_1 u$ in its integrability conditions), $\Gamma_T = -\eta E \int_0^T A_1 u(T-s,\cdot)(x_s^\eta)\, ds$; and if $L_0 V$ and $A_1 V$ are bounded and continuous and the martingale terms are square integrable, $\Gamma_T/T \to -\eta A_1 V(x_0)$ as $T \to 0$. The first variation of B-10 (c) holds as well if $A_1 u(T-s,\cdot)$ is bounded and continuous and $x_s^\eta \to x_s^0$ in probability for each s as $\eta \to 0$; that convergence is assumed here, not proved, because the noise amplitude now depends on η.

(c) Short-time sign near a minimum. Let x_* be a critical point of J with $D^2 J(x_*) \succ 0$ and $G_1(x_*) \succ 0$. For $V = J$, $A_1 J = -\langle \nabla J, G_1 \nabla J\rangle + \varepsilon \langle \mathrm{div}\, G_1, \nabla J\rangle + \varepsilon\ \mathrm{Tr}(G_1 D^2 J)$, and the short-time rate $-\eta A_1 J(x)$ (the limit in (b) whenever its hypotheses hold) is < 0 for every start x in a neighbourhood of x_*. The same change of mobility without noise ($\Sigma \equiv 0$ in both arms, drift $(A - G_\eta)\nabla J$) has short-time rate $\eta \langle \nabla J, G_1 \nabla J\rangle(x)$, which is > 0 for every $x \neq x_*$ in a neighbourhood of x_*. Near a non-degenerate minimum, the pairing reverses the short-time sign of a lift that increases the mobility. Proof. At x_*, $-\eta A_1 J(x_*) = -\eta\varepsilon\ \mathrm{Tr}(G_1 D^2 J)(x_*) < 0$ by the trace lemma of U-05 (c); $A_1 J$ is continuous, so the sign persists nearby. Without noise both arms are C^1 paths from x, and the chain rule and continuity give $\Gamma_T/T \to -\eta\langle g, \nabla J\rangle(x)$ with $g = -G_1 \nabla J$; near x_*, $G_1 \succ 0$ by continuity and $\nabla J(x) = D^2 J(x_*)(x - x_*) + o(|x - x_*|) \neq 0$ for $x \neq x_*$. ∎

(d) Long run. By U-08 (a), applied to (A, G_η, ε) for each η, $e^{-J/\varepsilon}$ is infinitesimally invariant for every arm: the lift cannot change the candidate invariant law. If $Z = \int e^{-J/\varepsilon} < \infty$ and both arms converge in law to $\mu_J = e^{-J/\varepsilon}/Z$ from x_0 — not proved here; U-08 (b) claims neither uniqueness nor convergence — then $\Gamma_T \to 0$ as $T \to \infty$ for every bounded continuous V.

(e) Drift lifts on a paired base. For a lift ηg as in B-10 on the base (M_P), with $g \in C^1$ and $\eta > 0$, $e^{-J/\varepsilon}$ stays infinitesimally invariant iff $\mathrm{div}(e^{-J/\varepsilon} g) \equiv 0$: for $f \in C_c^\infty$, $\int (L_\eta f) e^{-J/\varepsilon} = \eta \int \langle g, \nabla f\rangle e^{-J/\varepsilon} = -\eta \int f\ \mathrm{div}(e^{-J/\varepsilon} g)$, by U-08 (a) and one integration by parts. Example: $g = A' \nabla J$ with A' antisymmetric, C^1, $\mathrm{div}_c A' = 0$ (U-08 (★)).

(f) Explicit instance. $n = 1$, $J = x^2/2$, $A = 0$, $G = 1$, $G_1 = 1$: arm η is the Ornstein–Uhlenbeck toy of U-15 with $k = 1$, $\gamma = 1 + \eta$ and $\sigma = (2\varepsilon(1+\eta))^{1/2}$; deterministic x_0. By U-15 (b), $E\, x_T^\eta = x_0\, e^{-(1+\eta)T}$ and $E\,(x_T^\eta)^2 = \varepsilon + (x_0^2 - \varepsilon)\, e^{-2(1+\eta)T}$. (f1) $V = J$: $\Gamma_T = \tfrac{1}{2}(x_0^2 - \varepsilon)(e^{-2T} - e^{-2(1+\eta)T})$. For every $T > 0$ its sign is that of $x_0^2 - \varepsilon$: the lift helps above the noise level and hurts below it; $\Gamma_T \to 0$ as $T \to \infty$. Without noise ($\sigma \equiv 0$, same lift), $\Gamma_T = \tfrac{1}{2} x_0^2 (e^{-2T} - e^{-2(1+\eta)T}) \geq 0$. (f2) $V(x) = x^2 - x/2$, $x_0 = 1$, $\varepsilon = \tfrac{1}{2}$, $\eta = 1$: $\Gamma_T = \tfrac{1}{2}(2e^{-2T} - e^{-T} - e^{-4T})$, which is > 0 for $0 < T < \ln((1+\sqrt{5})/2)$ and < 0 for $T > \ln((1+\sqrt{5})/2)$. As in GTPA-C0113, the initial sign does not fix the sign at later horizons, and this persists under the pairing. Here the analogue of (K) in (b) holds: u is a polynomial in x with smooth coefficients in t (for $V = J$, $u(t,x) = \tfrac{1}{2}(x^2 e^{-2t} + \varepsilon(1 - e^{-2t}))$), it solves $\partial_t u = L_0 u$ by direct differentiation, and the integrability conditions need only the second moments above. Proof. (f1) From the moments. (f2) $E\, V(x_T^\eta) = \tfrac{1}{2} + \tfrac{1}{2} e^{-2(1+\eta)T} - \tfrac{1}{2} e^{-(1+\eta)T}$; subtract the two arms. With $z = e^{-T} \in (0, 1)$, $\Gamma_T = -\tfrac{1}{2} z(z-1)(z^2 + z - 1)$, whose sign is that of $z^2 + z - 1$, positive iff $z > (\sqrt{5} - 1)/2$. ∎

(g) Uniform scaling. If $A = 0$, $G_1 = G$ and $\Sigma_\eta = (1+\eta)^{1/2}\, \Sigma_0$, then $G_\eta = (1+\eta)G$ and $L_\eta = (1+\eta)\, L_0$, and the lifted arm is the base arm with time rescaled by $1+\eta$: by U-13 (c), whose proof applies verbatim to the paired drift b_P (it uses only that the whole drift is multiplied by the constant and Σ by its square root), x^η has the law of $t \mapsto x^0_{(1+\eta)t}$. So $\Gamma_T = E\, V(x_T^0) - E\, V(x^0_{(1+\eta)T})$: the efficacy of such a lift is the change of $E\, V$ along the base between T and $(1+\eta)T$. The instance (f) is of this kind.

En llano: si la acción aumenta la movilidad y el ruido sigue emparejado a ella, también aumenta el ruido: cerca del fondo de un valle la acción empeora al principio lo que sin ruido mejoraría, a la larga no cambia el reparto de equilibrio (si los dos brazos llegan a él), y el signo del efecto puede cambiar con el horizonte.

3. Block B — the eight answers

3.0 Summary

Row	Question (one line)	Verdict	Bearing statements
B-1 · EG-02.2	What `Hessian(Pi)` and `Covariance_Noise` are; which contraction	`ILL_POSED_AS_WRITTEN`	B-01; U-02, U-05, U-08
B-2 · EG-02.4	One typed object M_{cog}? On Q or on M_model? What is its cube?	`ILL_POSED_AS_WRITTEN`	B-07; U-12
B-3 · T6	Is Def 1.1's "rigidity" of the terrain the M_{cog} of Ax 6.1?	`ILL_POSED_AS_WRITTEN`	B-07; U-12
B-4 · EG-02.5	Which temperature; a pairing out of equilibrium; Γ versus G	`ILL_POSED_AS_WRITTEN`	B-09; U-08, U-11, U-15
B-5 · EG-03.7	Thm 6.1: the Itô bias of $\Pi(q)$ and where its compensation goes	`FAILS_WITH_COUNTEREXAMPLE`	B-01, B-05, B-06; U-02, U-14
B-6 · EG-04.1	"At resonance the impedance drops to zero"	`FAILS_WITH_COUNTEREXAMPLE`	B-08; U-11
B-7 · EG-09.3	Stochastic signed efficacy; what the pairing changes	`HOLDS_UNDER_STATED_HYPOTHESES`	B-10, B-11; U-01, U-02, U-05, U-08, U-13, U-15
B-8 · EG-10.b2	Does non-invariance under recoding single out Thm 6.1's term? Descent	`FAILS_WITH_COUNTEREXAMPLE`	B-02, B-03, B-04

3.1 B-1 · EG-02.2 — the notation of Correction_Ito

Question (triage `d0cf83e`, exact next question): "Are D^2 Pi / Hessian(Pi) and Cov_Noise / Covariance_Noise the same typed objects? What is Covariance_Noise as a function of Sigma (Sigma*Sigma^T, per unit time or not)? Which index contraction does Trace(...) denote for Pi: M_base -> M_model?"

GTPA v1.2 locators:

- p.7, summary 5.2: "Correction: Correction_Ito = - 0.5 * Trace(D^2 Pi * Cov_Noise)".
- p.38, §6.2 "Geometric Drift (The Ito-Stratonovich Correction)", Thm 6.1: "Correction_Ito = - 0.5 * Trace(Hessian(Pi) * Covariance_Noise)"; the sign was confirmed on the rendered page (MATH receipt T7).
- p.39: "Hessian(Pi): The tensor of second derivatives of the perception map (measuring how curved the lens is)".
- p.22, §2.3: "Pi: M_base -> M_model"; Ax 2.2 (a diffeomorphism onto its image).

Verdict: **`ILL_POSED_AS_WRITTEN`**.

Argument.

1. "D^2 Pi" (p.7) and "Hessian(Pi)" (p.38, glossed on p.39 as "the tensor of second derivatives") name one object, the second derivative of Π; for $\Pi : R^n \to R^d$ it has components $\partial_i \partial_j \Pi^a$. The p.7 line restates Thm 6.1's formula.
2. "Cov_Noise" and "Covariance_Noise" occur only on pp.7 and 38 and are never written in terms of Σ (triage). v1.2 does not say whether the covariance is $\Sigma\Sigma^\top$, $\Sigma\Sigma^\top \Delta t$ or something else, and does not write the contraction. As written, the expression has no single typed meaning.

3. What the base motion fixes (B-01 (a)): along the Itô motion the second-order drift of $\Pi(q)$ is $\frac{1}{2}\,\mathrm{Tr}_{\Sigma\Sigma^\intercal} D^2\Pi$, with $(\mathrm{Tr}_{\Sigma\Sigma^\intercal} D^2\Pi)^a = \sum_{i,j}(\Sigma\Sigma^\intercal)_{ij}\,\partial_i\partial_j\Pi^a$. So the reading under which Thm 6.1's expression is (minus) a drift is: covariance $= \Sigma\Sigma^\intercal$, the covariance of the noise increment per unit of time (a covariance over a step Δt would make it a displacement per step, not a rate); contraction = both base indices against $\Sigma\Sigma^\intercal$, model index free (§1). Under (P), $\Sigma\Sigma^\intercal = 2\varepsilon G$ (U-08).
4. The label: the term is the Itô second-order term of Π, not an Itô–Stratonovich conversion term. It is not in the written drift of the Stratonovich form of the image (B-01 (b); there it is carried by the Stratonovich integral), and it is built from $D^2\Pi$, not from $D\Sigma$ (UFD §3.2, item 3).

Repair adopted: GTPA v2 defines, for the base motion (M)/(M_P) and a C^2 perception map $\Pi : R^n \to R^d$, the Itô term of Π as $\frac{1}{2}\,\mathrm{Tr}_{\Sigma\Sigma^\intercal} D^2\Pi$ (componentwise as above; $\Sigma\Sigma^\intercal = 2\varepsilon G$ under (P)). $D^2\Pi$ and Hessian(Π) are one symbol; "Covariance_Noise" is replaced by $\Sigma\Sigma^\intercal$; the name "Itô–Stratonovich correction" is not used for this term. Where the term enters is EG-03.7.

En llano: GTPA escribe el mismo término con dos notaciones y nunca dice qué es la "covarianza del ruido" ni cómo se contraen los índices; la nueva edición lo fija: es $\Sigma\Sigma^\intercal$ (la covarianza del ruido por unidad de tiempo), contraída con las dos derivadas de la lente, y el término se llama "término de Itô de la lente".

3.2 B-2 · EG-02.4 — the three variants of M_cog

Question: "Is there one typed object M_cog of which (a) matrix, (b) spectrum and (c) scalar power are well-defined functions? Is it defined on Q (Hessian of J) or on M_model (internal potential)? Does 'the cube of the cognitive mass' mean det, trace, operator norm or dim N?"

GTPA v1.2 locators:

- (a) matrix: p.5, summary 1.2, "M_cog(q) := Sqrt((D^2 J(q))^2 + epsilon * I)"; p.15, Ax 0.1, "M_cog(q) := Sqrt(D^2 J(q)^2 + epsilon * I)"; p.18, Def 1.1, "M_cog(q) := Sqrt(H(q)^2 + epsilon_inertial * I)", followed by "'Cognitive Mass' in an Agent is physically realized as the rigidity of its internal manifold M_model against deformation".
- (b) spectrum: p.64, App A.2, "Mass is the Eigenvalue Spectrum of the Hessian H(q) = D^2 J(q)"; p.81, App F.1, "the spectrum of the Hessian operator of the internal potential ... Let N = dim(M_model) ... the set of N eigenvalues {lambda_1, ..., lambda_N} of the Hessian H".
- (c) cube: p.21, Thm 2.1, and p.85, Cor F, "C_res ~ (M_cog)^3"; p.6, summary 2.1; p.31, "Complexity Penalty ... (M_cog^3)"; derivations pp.21–22 and pp.82–85.
- Uses as a matrix: p.34, Def 5.1 (`M_cog^(-1)`); p.36, Thm 5.1, condition 2; p.37, Ax 6.1, "dq_est/dt = M_cog^(-1) * p_est"; p.38, component 3.

Verdict: **`ILL_POSED_AS_WRITTEN`**.

Argument.

1. (a) is a well-defined matrix: for the symmetric Hessian H and $\delta > 0$, $(H^2 + \delta I)^{1/2}$ is the unique symmetric positive definite square root, with eigenvalues $(\lambda_i^2 + \delta)^{1/2}$ (B-07 (a)). It is always invertible, so Ax 6.1's inverse exists and Thm 5.1's condition 2 holds automatically (B-07 (b)).
2. (b) is not a function of (a), and (a) is not a function of (b) (B-07 (d)): (a) forgets the signs of the eigenvalues (B-07 (c)), while App F's "set of eigenvalues of the Hessian" is signed. Both are functions of the Hessian H itself. So the one typed object behind (a) and (b) is H, not "M_cog".
3. (c): in both printed derivations the cube comes from the dimension: "N_ops ~ (4/3) * N^3" (p.83), "N * N * N = N^3" (p.84), "N_max ~ (P_budget / (beta * omega))^(1/3)" (p.85), and

the $N^1/N^2/N^3$ count of Thm 2.1 (pp.21–22); p.22 also says "A linear increase in M_cog yields a cubic increase in C_res". None of them uses a determinant, a trace or a norm of the matrix (a). So "the cube of the cognitive mass" means N^3 with $N = \dim M_{\text{model}}$ — the reading its printed derivations support — a function of the size of the matrix, not of its entries, hence not of the curvature. (On p.21 C_{res} is a capacity; on p.85 the same display follows "Cost scales with the cube of Resolution".) Whether these operation counts are necessary is not asked here.

4. Q or M_model: v1.2 writes both. Def 1.1's formula uses $H(q) = D^2J(q)$, the Hessian of J on Q ($n \times n$), while its implication sentence, App F ($N = \dim M_{\text{model}}$) and Ax 6.1 (the q_{est} equation on M_model, $d \times d$) put M_{cog} on the model side. A text sweep of "M_cog" (pp.3, 5–8, 10, 15, 18, 21–22, 31, 34, 36–38, 48–49, 81, 85) finds no map from the Q-side Hessian to the model side, and for $d \neq n$ no identification of an $n \times n$ matrix with a $d \times d$ matrix exists. So "on Q or on M_model" has no answer inside v1.2.
5. The name: UFD's curvature of J is "rigidez" and nothing else — card F05, "rigidez, no masa" (U-12 (e)). A GTPA object called a mass cannot be imported from UFD's curvature.

Repair adopted (the architecture's default: re-axiomatize the import as a GTPA parameter): in GTPA v2, M_{cog} is a declared symmetric positive definite $d \times d$ matrix field on the model state space — the matrix that Ax 6.1 inverts. It is not derived from UFD and it is not called a mass. If a GTPA model ties it to a curvature, it declares the map (for instance $M_{\text{cog}} \coloneqq (H^2_{\text{model}} + \delta I)^{1/2}$ for a declared model-side Hessian H_{model} and $\delta > 0$) as a modeling choice, with B-07's caveats. The cubic statements, if kept, are written in terms of $N = \dim M_{\text{model}}$ (their status is Block C). UFD's curvature (U-12) stays in UFD, with no mass reading; the regularizer is renamed δ (ε is UFD's noise level).

En llano: GTPA llama "masa cognitiva" a tres cosas distintas (una matriz, un espectro con signo y, en la ley cúbica, en realidad la dimensión N) y en dos espacios distintos; la nueva edición la deja como un parámetro declarado de la dinámica de creencias, sin llamarla masa ni sacarla de la curvatura de UFD, que es "rigidez" y nada más.

3.3 B-3 · T6 — the INERCIA caution and M_cog (follows EG-02.4)

Question. Demand: "Which EG-02 row (if any) covers the INERCIA caution 'una secuencia barata no prueba disciplina ni su falta; describe una forma del terreno' (Def 1.1, p.18)?" Adjudicated outcome (triage, reviewer adopted): covered by EG-02.4; "the doubt behind the caution is whether Def 1.1's 'rigidity of the terrain' (M_cog on H(q), p.18) is the same object as the M_cog of Ax 6.1 (p.37, on M_model)"; "The plain-language caution ... is FORMACION's nonprobative gloss and needs no GTPA row."

GTPA v1.2 locators: p.18, Def 1.1 (formula on $H(q)$; "Valley Regime (H >> 0): The vacuum is 'stiff' and stable"; "Saddle Regime (H << 0): The vacuum is 'stiff' but unstable"; the implication sentence on M_model); p.37, Ax 6.1. The words "cheap", "discipline" and "sequence" do not occur in Def 1.1 (triage); the caution is FORMACION's gloss (quasar `FICHAS-CORE-INERCIA-01.md` l.476–478, per the triage reviewer).

Verdict: **`ILL_POSED_AS_WRITTEN`**.

Argument.

1. As in EG-02.4, item 4: Def 1.1's object lives on Q, Ax 6.1's on M_model, and v1.2 gives no map between them, so "the same object?" has no answer in v1.2.
2. What carries "a shape of the terrain" is the signed curvature of J: valley, saddle or flat directions (U-12 (b)–(d)). Def 1.1's M_{cog} erases exactly that sign (B-07 (c)); its valley and saddle regimes give the same matrix.

3. The caution maps a written sequence to a terrain shape. No formal object or observation map for that exists in v1.2 or in the lab catalog (triage); this stays as the triage left it.

Repair adopted: in GTPA v2 the two are different objects by declaration. “La forma del terreno” is the curvature of J (U-12, “rigidez”), a UFD object; M_{cog} is the declared GTPA matrix of EG-02.4’s repair. FORMACION’s caution stays a declared procedure without a GTPA bridge; a bridge “written sequence → M_cog / terrain shape” would be `UNMAPPED_TO_GTPA` unless an exact object and an observation map are supplied.

En llano: la “forma del terreno” de la ficha es la curvatura de J (la rigidez de UFD), y el M_cog de la dinámica de creencias es otro objeto; GTPA v1.2 los llama igual sin decir cómo se pasa de uno a otro, y la nueva edición los separa.

3.4 B-4 · EG-02.5 — which temperature, out of equilibrium, Γ versus G

Question: “Is T in Ax 0.3 the same variable as T_info(t) or T_sub? Can an FDT hold instantaneously while T_info evolves by $d(T_{\text{info}})/dt = k^*_{\text{heat}}(E_{\text{Hamiltonian}} - E_{\text{expected}}) - k^*_{\text{cool}}(T_{\text{info}} - T_{\text{min}})$ (p.39), i.e. out of equilibrium? How do Gamma (a friction on p) and the mobility G(q) of Ax 1.2 relate?”

GTPA v1.2 locators:

- p.16, Ax 0.3: “Sigma(q) * Sigma(q)^T = 2 * Gamma * k_B * T” (“Origin: TECD Chapter 29”).
- p.31, Thm 4.1, and p.7, summary 4.3: “Sigma(t) * Sigma(t)^T = 2 * Gamma(t) * k_B * T_info(t)” (“Inherited from TECD Chapter 29”).
- p.39, Control Law 6.1: “The Informational Temperature T_info (which drives the internal noise variance)” and the displayed law.
- p.29, Def 4.1: “T_substrate: The effective operating temperature of the hardware”; p.67, App B.1 (“T_sub”); p.87, App G.3: “sigma_thermal^2 ~ k_B * T_sub”.
- p.17, Ax 1.2 (mobility $G(q)$); pp.43, 67, 70–71 (`Gamma` multiplying p).

Verdict: **`ILL_POSED_AS_WRITTEN`**.

Argument.

1. v1.2 never identifies T, T_{info} and T_{sub} with one another, and it attaches them to different noises: T in Ax 0.3 to the territory’s $\Sigma(q)$, the noise of Ax 1.2; $T_{\text{info}}(t)$ to the agent’s exploration noise (“Creativity and exploration require the injection of stochastic noise (Sigma) into the control loop”, p.31; “the internal noise variance”, p.39); T_{sub} to “the hardware”, to `beta` (pp.29, 67) and to App G’s `sigma_thermal` (p.87). “Is T the same variable as T_info or T_sub” has no answer inside v1.2. (UFD §3.5, item 1, already shows that Ax 0.3 and Thm 4.1 are two different relations under one name.)
2. Out of equilibrium. As a constraint on coefficients, a pairing can be imposed at every instant for a level that changes in time; it only defines the noise at that instant. What does not follow is the equilibrium reading: in B-09 (one dimension, pairing at every time) the law is the instantaneous law $\propto e^{-J/\varepsilon(t)}$ at all times only if ε is constant, and when the level moves the law lags behind. With T_{info} driven by the state (Control Law 6.1), the level becomes a process coupled to the state; that joint process is outside the UFD base (C4 and C6: one constant ε, time-independent coefficients) and is not studied here.
3. Γ and G. v1.2 states no relation. The relation mathematics supplies is a limit (U-11 (b), (c); UFD §3.5, item 3): for a (q,p) system whose coefficient of $v = M^{-1}p$ is γ, the small-m limit has mobility $G = \gamma^{-1}$ ($G = (m\Gamma)^{-1}$ when Γ multiplies p and $M = mI$; the limit is taken with $\gamma = m\Gamma$

fixed, U-11 (b)), and with a q-dependent coefficient the limit is the paired motion (M_P) with $A = 0$ (in one dimension, U-11 (c); the multi-dimensional case is by citation only).

Repair adopted: GTPA v2 has no temperature (physics CLOSED; k_B, T, T_{sub} are not objects). The base carries one constant noise level ε with the pairing (P) (C4, U-08). If GTPA v2 keeps a time-varying level for the agent's exploration noise, it declares it as a GTPA schedule $\varepsilon_{\text{cog}}(t)$ — not UFD's ε, and not a temperature — with its pairing as a constraint at each instant and no equilibrium statement drawn from it (B-09). Γ versus G is U-11.

En llano: GTPA usa tres "temperaturas" (T, T_info, T_sub) para ruidos distintos sin decir si son la misma; la nueva edición no tiene temperatura: la base tiene un solo nivel de ruido fijo ε, y si un nivel cambia con el tiempo se puede emparejar en cada instante, pero el reparto va con retraso respecto del "equilibrio del instante"; Γ y G se relacionan por el límite de UFD (G es la inversa del coeficiente de p), no son el mismo coeficiente.

3.5 B-5 · EG-03.7 — Thm 6.1: prove, degrade or bound

Question: "For q_est = Pi(q) with q following Ax 1.2, does Itô's lemma give a drift bias of +0.5*Tr(Hessian(Pi) Sigma Sigma^T), so that the compensating term is exactly -0.5*Tr(...)? Why does it enter dp_est/dt (a momentum equation) rather than dq_est/dt?"

GTPA v1.2 locators:

- p.38, §6.2: "When a stochastic process is mapped through a non-linear curved lens (the Perception Map Pi), the mean of the distribution shifts"; "The Geometric Bias Problem: Mean(Pi(Noise)) != Pi(Mean(Noise))".
- p.38, Thm 6.1: "To maintain an accurate representation of the centroid of probability, the evolution of the momentum p_est must include a compensatory force derived from the curvature of the perception map. Correction_Ito = - 0.5 * Trace(Hessian(Pi) * Covariance_ Noise)". No proof block (inventory).
- p.37, Ax 6.1: "dq_est/dt = M_cog^(-1) * p_est dp_est/dt = F_cognitive - Gamma_cog * p_est + Correction_Ito"; p.39 (gloss); p.7, summary 5.2.

Verdict: **`FAILS_WITH_COUNTEREXAMPLE`** (Thm 6.1 as printed: the term in the p_{est} equation does not compensate the bias; the Itô identity behind it holds).

Argument.

1. The bias. On the UFD base (Itô, C2), with EG-02.2's typing, Itô's formula gives exactly: Itô drift of $\Pi(q) = D\Pi\, b + \tfrac{1}{2}\,\mathrm{Tr}_{\Sigma\Sigma^{\mathsf{T}}}\, D^2\Pi$ (B-01 (a)). So yes: the bias is $+\tfrac{1}{2}\,\mathrm{Tr}(D^2\Pi\,\Sigma\Sigma^{\mathsf{T}})$, and $-\tfrac{1}{2}\,\mathrm{Tr}(...)$ removes it exactly from the drift of the image, leaving the push-forward $D\Pi\, b$. In B-06 it is also exactly the rate at which "Mean(Pi(Noise))" leaves "Pi(Mean(Noise))". If Ax 1.2's noise is read in the Stratonovich sense (v1.2's declaration), the second-order term is not in the written drift of the Stratonovich form of the image (B-01 (b)). It is carried by the Stratonovich integral $\int D\Pi\Sigma \circ dW$, whose mean is in general not zero (B-01 (b), proof). So the rate of $E[\Pi(q_t)]$ contains $\tfrac{1}{2}\,\mathrm{Tr}_{\Sigma\Sigma^{\mathsf{T}}}\, D^2\Pi$ under either reading, and in B-06 (constant Σ, where the two readings give the same law) the mean gap $E[\Pi(q_t)] - \Pi(E\,q_t)$ is the same under both. The convention must be declared because it changes the base Itô drift by c_Σ (U-02; U-14 (d) shows what a silent swap does).
2. Placement. The bias is a rate of the image coordinate, so it belongs to the q_{est} equation. Ax 6.1 has no drift slot there ($dq_{\text{est}}/dt = M_{\text{cog}}^{-1}\, p_{\text{est}}$) and puts the term in the p_{est} equation, where it acts one order later in time: its contribution to q_{est} is $\tfrac{1}{2} M_{\text{cog}}^{-1}\, \varkappa\, t^2 + O(t^3)$, and in the long run

it offsets a rate bias only if $\Gamma_{\text{cog}}\, M_{\text{cog}} = I$ (B-05). v1.2 gives no derivation of the placement (no proof block), so "why dp_est/dt" has no answer in the text.

3. Counterexample (B-06): flat J, the pairing, the diffeomorphism $\Pi(q) = (q_1, q_2 + q_1^2/2)$ of R^2 (so Ax 2.2 holds), constant M_{cog} and Γ_{cog}. The perceived mean drifts away from the image of the mean at the constant rate $\bar{\kappa} = (0, \varepsilon)$, while Thm 6.1's term in the p_{est} equation moves the estimate by $-\tfrac{1}{2} M_{\text{cog}}^{-1}\, \bar{\kappa}\, t^2 + O(t^3)$, which does not remove a bias $\bar{\kappa}\, t$ at small t, whatever M_{cog}. So including the printed term does not "maintain an accurate representation of the centroid of probability" at first order in time, in either reading of the centroid ($E[\Pi(q_t)]$ or $\Pi(E[q_t])$).

4. What the correction produces. Removing the term leaves the push-forward $D\Pi\, b$, the part of the image drift that transforms as a vector under a recoding of the model side (B-03 (a), B-04 (a)). The perceived mean $E[\Pi(q_t)]$ itself moves with the term (B-01 (a)), so a correction moves an estimate away from it: which quantity q_{est} is meant to track must be declared.

Repair adopted: GTPA v2 states the identity B-01 (a), with the Itô term of the lens typed as in EG-02.2. If it keeps a correction, it puts it in the equation of the image coordinate, as a rate ($dq_{\text{est}}/dt = \ldots - \tfrac{1}{2}\, \text{Tr}_{\Sigma\Sigma^{\mathsf{T}}}\, D^2\Pi$, evaluated where the model declares), and it declares what q_{est} tracks. The claim that the printed term keeps an accurate centroid is withdrawn.

En llano: la cuenta de Itô que hay detrás es correcta (la curvatura de la lente añade ½·Tr($\Sigma\Sigma^{\mathrm{T}}$·D²Π) a la deriva de lo que se ve), pero GTPA pone la corrección en la ecuación de p_est, donde actúa tarde y no la compensa; la nueva edición la pone, si la usa, en la ecuación de q_est y dice qué se quiere seguir.

3.6 B-6 · EG-04.1 — "at resonance the impedance drops to zero"

Question: "Is there a frequency-domain impedance Z(omega) derivable from the linearized dynamics (Hessian H(q), Gamma, M_cog) whose magnitude is minimized at omega_ctrl = omega_base? Is its minimum zero or the resistive (damping) part? How does it relate to Def 7.2's Z?"

GTPA v1.2 locators:

- p.46, §7.5, "Physical Effect", item 2: "Impedance Drop: At resonance, the mechanical impedance of the load drops to zero (pure resistance). The force u required to achieve a displacement dq is minimized."
- p.46, Def 7.3: "a characteristic frequency spectrum omega_base derived from the eigenvalues of the Hessian H(q)"; the tuning rule "min_omega || omega_ctrl - omega_base ||".
- pp.43–44, Def 7.2: "Z := Hessian_u(Action_Local) Z = B^T * G_reg * B + lambda_effort * I" (summary 6.1, p.8). Locator correction (triage): Z is §7.2 / Def 7.2, not §7.1.
- Related: p.26 ("Truth as Low Impedance"); p.58, Def 10.4.

Verdict: **`FAILS_WITH_COUNTEREXAMPLE`**.

Argument.

1. For a linear second-order motion — the shape of v1.2's (q,p) forms, linearized at a minimum — the ratio of forcing to time-derivative response is, mode by mode, $R(\omega) = c + i(m\omega - k/\omega)$; its modulus is minimized exactly at the mode's own frequency, where it equals the coefficient c of the first derivative (B-08 (a), (b)). "Drops to zero" and "(pure resistance)" contradict each other unless $c = 0$. Counterexample: $m = k = c = 1$, $|R(1)| = 1$. With several modes and $C = \gamma M$, each mode j has its minimum γ at its own ω_j, and the ω_j^2 are the eigenvalues of $M^{-1}H$: a frequency needs the coefficient matrix of the second derivative as well as H (Def 7.3 names only H; EG-04.2, Block C).

2. v1.2's second sentence ("The force u required to achieve a displacement dq is minimized") is the ratio $K(\omega) = F/X$: with $c > 0$ it is minimized at $\omega_* < \omega_0$, or at no positive frequency when $c^2 \geq 2mk$, and its minimum is positive (B-08 (a)). So the displacement reading does not give $\omega_{\text{ctrl}} = \omega_{\text{base}}$ either, unless $c = 0$.
3. On the UFD base the answer changes: the first-order motion (C5) has no second-order structure; its gradient part, linearized at a non-degenerate minimum, has no preferred positive frequency in either ratio; a circulating part can create one, at its rotation rate, and there the minimum is the sliding coefficient, not zero (B-08 (c)). (q,p) systems are separate objects (U-11 (a)).
4. Def 7.2's Z is a constant matrix, the Hessian in u of the static quadratic `Action_Local`. It has no frequency argument, and with $G_{\text{reg}} \succeq 0$, $\lambda_{\text{effort}} > 0$ it satisfies $Z \succeq \lambda_{\text{effort}} I$, so it cannot drop to zero (B-08 (d)). It is a different object with the same name.

Repair adopted: GTPA v2 replaces the sentence by B-08: at a mode's own frequency the reactive part of the ratio vanishes and what remains is the coefficient of the first derivative (zero only when that coefficient is zero); a frequency needs both the stiffness matrix and the coefficient matrix of the second derivative (EG-04.2); the first-order base has no such tuning unless a circulating part is present; Def 7.2's Z is kept as the Hessian of the control objective and is not identified with any frequency response.

En llano: en un sistema que oscila, a la frecuencia propia no desaparece todo: desaparece la parte "reactiva" y queda la amortiguación, que solo es cero si no hay amortiguación; en el movimiento de primer orden de UFD no hay frecuencia preferida salvo que haya giro; y la Z de la Def 7.2 es otra cosa, una matriz fija que nunca es cero.

3.7 B-7 · EG-09.3 — a stochastic efficacy lane

Question: "Take a declared finite-dimensional Stratonovich SDE dx = (f0 + eta*g)(x) dt + Sigma(x) circ dW on a compact tube, with exact eta=0 ablation and a fixed C2 observable V. Does the expected signed efficacy Gamma = E[V(x_0(T))] - E[V(x_eta(T))] have an exact representation through the backward Kolmogorov/adjoint equation, with first variation, short-time limit and representation-transport invariance analogous to C0115 T2-T6? Which clauses fail, or change sign behaviour (C0113-type reversals), when Sigma Sigma^T is tied to a dissipation coefficient by an FDT relation?" (Triage reviewer corrections adopted: "T2-T6" reads T2–T4 and T6; the FDT clause depends on EG-09.1/09.2; the ledger's MATH-transfer records are not inheritable.)

Locators: no single v1.2 object. v1.2: Ax 1.2, p.17 (the base SDE, read there in the Stratonovich sense); Ax 0.3, p.16, and Thm 4.1, p.31 (the "FDT" relations). Lab: GTPA-C0115 (T1–T4, T6), GTPA-C0112 and GTPA-C0113 (bounded semigroup bridge; sign frontier); the N0–N3 interface of `GTPA-MR-T1-TYPED-BINDING-SOUNDNESS-PREREG-01.md` §4.

Verdict: **`HOLDS_UNDER_STATED_HYPOTHESES`**.

Argument.

1. Base. EG-09.1 and EG-09.2 are settled by UFD's convention: the base is the Itô motion (M) or, under (P), (M_P) on R^n (C1–C5). Stratonovich input with an η-independent Σ converts with a conversion term common to both arms, so the ablation stays exact (B-10 (f)).
2. What holds, for a lift in the drift: exact ablation (B-10 (a)); an exact representation through the backward Kolmogorov equation, (★) (B-10 (b), under the named hypothesis (K)) — the stochastic counterpart of the deterministic identity behind C0115's adjoint formula, and a Duhamel formula with an unbounded perturbation, outside C0112's bounded class; an L^2 bound and the first variation (B-10 (c), globally Lipschitz coefficients, bounded g); the short-time limit $-\eta\langle g, \nabla V\rangle(x_0)$, the same as without noise (B-10 (d)); invariance of Γ_T under C^3 changes of coordinates, with transported lift direction $\varphi_* g$ (B-10 (e)).

3. What fails: the compact tube. For Brownian motion (B-10 (g)) the paths leave every compact set with positive probability, so a compact tube cannot be assumed in general; global hypotheses on R^n replace it. (Nonzero noise alone does not force this: noise that vanishes at the boundary can keep the paths in a compact set, B-10 (g).)
4. What the pairing changes (the FDT relation of the new edition is (P), U-08). If the lift acts on the drift only, the noise is common to both arms and (P) plays no role in items 2–3; the lift keeps $e^{-J/\varepsilon}$ infinitesimally invariant iff $\operatorname{div}(e^{-J/\varepsilon} g) \equiv 0$ (B-11 (e)). If the lift acts on the paired mobility, the noise moves with it: the lift operator becomes second order (B-11 (a)); representation, first variation (with continuity in η assumed) and short-time limit still hold with that operator (B-11 (b)); the short-time sign reverses near a non-degenerate minimum — for $V = J$ a mobility increase has short-time rate < 0 there, while without noise it is > 0 (B-11 (c)); the long-run efficacy, for bounded continuous V, is zero whenever both arms converge to their common invariant law, because under (P) that law does not depend on the mobility (B-11 (d)); in the explicit Ornstein–Uhlenbeck pair the sign of Γ_T for $V = J$ is that of $x_0^2 - \varepsilon$ at every horizon (B-11 (f1)), and a uniform paired speed-up only rescales time (B-11 (g)).
5. C0113-type reversals persist under the pairing: in the same explicit pair, $V(x) = x^2 - x/2$ gives $\Gamma_T > 0$ before $T = \ln((1+\sqrt{5})/2)$ and $\Gamma_T < 0$ after (B-11 (f2)).
6. Lab reading (no binding). In MR-T1's frozen interface this is N1 with $A_{\text{act}} = \langle g, \nabla\cdot\rangle$ or A_1 on an N0 base, and Γ_T is N3's Γ at x_0. Filling N0 from UFD needs the written UFD→GTPA N0 record, with the seven MR-T1.N0 obligations; nothing here discharges them. The ledger's diffusion-transfer records that wait on a MATH release (GTPA-MC-T01/T02, GTPA-MAP-M2-PHI-04, GTPA-CS-C05, as listed by the triage reviewer) are not used and not inheritable (MATH = UNBOUND). B-10 and B-11 become ledger claims only through the gate, from C0138; the triage's "viability and cost" question reduces to that gate.

Repair adopted: none needed for the verdict; the new edition states B-10 and B-11 with their hypotheses, and drops the compact tube.

En llano: sí: con ruido, la eficacia de una acción se escribe exactamente, con su primera variación, su límite a tiempos cortos y su independencia de las coordenadas, bajo las hipótesis escritas; falla el "tubo compacto" del caso sin ruido, que ya no se puede suponer en general porque el ruido puede sacar las trayectorias de cualquier compacto; y si la acción cambia la movilidad con el ruido emparejado, cambia también el ruido, lo que invierte el signo al principio cerca del fondo de un valle y anula el efecto a la larga cuando los dos brazos llegan al equilibrio.

3.8 B-8 · EG-10.b2 — "non-invariant under change of representation implies curvature artifact"

Question: "Fix a noise law on M_base, a C2 perception map Pi and diffeomorphisms phi of M_model. Is the component of Mean(Pi(noise)) - Pi(Mean(noise)) that changes under Pi -> phi o Pi exactly the -0.5 Trace(Hessian(Pi) Cov_Noise) term of Thm 6.1, to the declared order in the noise? Under which equivalence does the corrected centroid descend in the sense of C0121?"

Locators: pp.38–39 (Thm 6.1; "Mean(Pi(Noise)) != Pi(Mean(Noise))"; "measuring how curved the lens is"); p.79, Thm E.1 ("Modulo diffeomorphisms", with `Pi` absorbing a Hessian term), context only; ledger GTPA-C0121 (descent to a quotient). Sign confirmed on the rendered p.38 (T7).

Verdict: **`FAILS_WITH_COUNTEREXAMPLE`**.

Argument.

1. Order. To second order in the noise, $\text{Mean}(\Pi(\xi)) - \Pi(\text{Mean}\ \xi) = +\tfrac{1}{2}\ \text{Tr}_C\, D^2\Pi(m)$, with a remainder bounded by the third moment (B-02). So the whole gap is, to that order, Thm 6.1's

expression with the opposite sign — for every presentation, $\varphi \circ \Pi$ included.

2. Change under recoding. That expression is itself representation-dependent: $\mathrm{Tr}_C\, D^2(\varphi \circ \Pi) = D\varphi \cdot \mathrm{Tr}_C\, D^2\Pi + \mathrm{Tr}_{D\Pi\, C\, D\Pi^\intercal}\, D^2\varphi$ (B-03). So the part of the gap that does not transform as a vector under $\Pi \mapsto \varphi \circ \Pi$ is $\tfrac{1}{2}\, \mathrm{Tr}_{D\Pi\, C\, D\Pi^\intercal}\, D^2\varphi(\Pi(m))$: it comes from the curvature of the recoding φ, not from Hessian(Π). Counterexamples: (i) φ affine and $D^2\Pi \neq 0$: nothing changes beyond the vector law, while Thm 6.1's term is not zero; (ii) $\Pi = \mathrm{id}$, $\varphi(y) = y + y^3/3$, Gaussian ξ with mean $m \neq 0$ and variance C: Thm 6.1's term for Π is 0, while the gap of $\varphi \circ \Pi$ is exactly $mC \neq 0$. So non-invariance does not single out the Hessian(Π) term; the triage's rival reading ("the Ito-Stratonovich correction term is itself coordinate-dependent") holds.
3. Descent (C0121: a label descends iff it is constant on classes; for a point or a vector, iff it is equivariant). At the level of the motion, the corrected drift $D\Pi\, b$ descends exactly under all C^2 recodings of the model side, while the uncorrected image drift descends only under affine recodings (B-04 (a)). At the level of a fixed noise law, the corrected centroid $\mathrm{Mean}(\Pi(\xi)) - \tfrac{1}{2}\, \mathrm{Tr}_C\, D^2\Pi(m)$ descends exactly under affine recodings and, under C^3 recodings with bounded derivative, up to a third-order defect that can be nonzero (B-04 (b)); $\Pi(\mathrm{Mean}\ \xi)$ descends exactly under all recodings.

Repair adopted: GTPA v2 states B-02–B-04 in place of the informal principle: the gap is, to second order, the curvature term of the presentation in use; what the model-side coordinates cannot change is the corrected drift (exactly) and the image of the mean; the corrected centroid is coordinate-free exactly for affine recodings and, for curved ones, to second order in the noise.

En llano: lo que cambia al recodificar las coordenadas del modelo no es el término de Thm 6.1, sino un término que sale de la curvatura de la propia recodificación; lo que no depende de las coordenadas es la deriva corregida (exactamente) y la imagen de la media, y la media corregida solo lo es exactamente en cambios afines, y en cambios curvos salvo un error de tercer orden.

4. What this file does not say

- No physics: no temperature, mass, friction, force, momentum, impedance or resonance as a physical object; v1.2's names are used as names.
- No GTPA claim: nothing enters the ledger and no claim ID is assigned.
- No MATH validity: no MATH object is used; the ROADMAP 22 records (MATH→UFD, MATH→GTPA N0) are not written here; no binding is created; the MR-T1 N0 obligations are not discharged.
- No UFD statement is added or changed; M_{cog} is not a UFD object (card F05).
- Nothing about persons, minds or agents in the world; no empirical claim.
- Not answered here: the Block C questions (among them EG-04.2 and the status of Thm 2.1, F.1 and Cor F); the joint dynamics of the state and a state-driven noise level (Control Law 6.1); what q_{est} should track (declared by the model, EG-03.7 repair).

5. Open points (not asserted)

- Hypothesis (K) for given coefficients (B-10 (b), B-11 (b)).
- Continuity in η of the paired lift (B-11 (b), first variation).
- Convergence in law of both arms to μ_J (B-11 (d)).

- A compact state space without boundary for B-10 (instead of R^n).
- The evaluator read of this file.

6. Sources and references

Pins: research baseline `3c2052042aa16c7a07ea4be3a1d4bccc7428df45`; UFD nucleus `ufd/UFD-NUCLEO.md` (blob `4f52a62c`); triage `d0cf83e3f1e7ac50115509f32d00aa58ed0bd51a` (`handoffs/governance/2026-09-25-formacion-eg01-eg10-eg19-triage-01.md`, blob `00a1e1eb`; .json, blob `a772d5cf`); architecture proposal `handoffs/governance/2026-09-25-theory-architecture-proposal-v1.md` (blob `7fc2974c`); ledger `gtpa-lab/CLAIM-LEDGER.md` (blob `84b445d6`); MR-T1 preregistration and result (`gtpa-lab/source-audits/`, blobs `756b0bee`, `4d6f42a8`); GTPA v1.2 text `quasar:modulos/formacion/fuentes/GTPA-v1.2.txt` (blob 32787369) and PDF (SHA-256 `0c9434c1c6f687e1e16100929641a1a81de0a0c9ac808d18b6991eb9cf75865f`).

Textbooks (locators as used above):

- R. A. Horn, C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013 — Thm 7.2.6.
- I. Karatzas, S. E. Shreve, *Brownian Motion and Stochastic Calculus*, 2nd ed., Springer, 1991 — §3.2; Thm 3.3.6; Thm 5.2.9.
- B. Øksendal, *Stochastic Differential Equations*, 6th ed., Springer, 2003 — Cor. 3.1.7, Thm 3.2.1, Thm 4.2.1.
- The UFD nucleus's references (UFD §9) for the statements U-01..U-15 cited here.